

WEB PROGRAMMING - BCSE203E
ASSIGNMENT 7

NAME: AJAY BERLIN

REG NO: 24BCE0449

FACULTY: Nihaal Ahmed K

QUESTION

Design and Development of a ReactJS-Based Campus Assistance Hub

A university intends to develop a Campus Assistance Hub, a Single Page Application (SPA) that enables students to explore important campus services such as Academic Support, Library Services, Transportation, Hostel Services, Health & Wellness, and Technical Support through a unified web interface.

Design and develop a complete, responsive, and interactive Single Page Application using ReactJS. The application shall present the available campus services through a professionally designed interface and allow users to select or explore a service and dynamically view its relevant information without navigating to separate HTML webpages.

The application shall be developed using the ReactJS concepts covered in Module 6, including React Environment Setup, ReactJS Basics, JSX, React Components, React Component API, React Component Life Cycle, Constructors, and React Developer Tools. Students are expected to analyze the requirements and design an appropriate component-based architecture consisting of meaningful and reusable components.

Repeated elements such as service cards, categories, or information sections shall be implemented using reusable React components rather than independently duplicated structures. A suitable constructor shall be used for component initialization, and at least one appropriate component life-cycle method shall be demonstrated to perform or indicate an initialization or update-related operation.

The application shall provide meaningful interaction within the single page. For example, selecting a campus service may dynamically display its description, contact information, location, operating hours, or other relevant details. The implementation should demonstrate

how JSX, components, component structure, and life-cycle behaviour work together to create a maintainable web interface.

Students shall also use React Developer Tools to inspect the component hierarchy of their application and provide evidence of its usage. The final solution should demonstrate design thinking and component reusability, rather than simply reproducing a static webpage.

VIT Campus Context and Branding

The application shall be developed with VIT – Vellore Institute of Technology, Vellore Campus as the contextual reference. Students should consider realistic campus requirements, services, locations, facilities, and student-use scenarios while designing the application.

The application should appropriately incorporate:

- VIT logo / institutional branding
- VIT – Vellore Institute of Technology name
- VIT Vellore campus context
- Suitable campus-related visual elements or imagery
- A professional university-oriented interface

The information presented in the application may be sample/mock information for academic demonstration and need not represent official VIT service information.

HOTS / Design Analysis

Suppose the Campus Assistance Hub is expanded from 5 services to 50 services. Analyze how your component design would affect code reusability, duplication, and maintainability. Identify one important component-design decision made in your application and justify why it would remain useful as the application grows.

SCREENSHOTS:



Campus Assistance Hub
VIT VELLORE MAIN CAMPUS

LATEST NOTICE

 We

 **AJAY BERLIN**
248CE8449

 10:20:21 PM IST

 HOTS

 Emergency

VIT VELLORE

Official Student Services & Facility Portal

Unified Campus Assistance Hub

Explore and access academic advising, 6–tier central library resources, intra–campus shuttles, residential facilities, 24/7 emergency health care, and technical support — dynamically rendered within a single interactive interface.

8
CAMPUS SERVICES

24/7
EMERGENCY CARE

SPA
ZERO PAGE
RELOADS

M6
REACTJS LIFECYCLE

Campus Services Directory

Select any campus service card below to dynamically view comprehensive operational details in the SPA inspection panel.

8 of 8 Services Available



Campus Assistance Hub
VIT VELLORE MAIN CAMPUS

LATEST NOTICE

 Welcom

 **AJAY BERLIN**
248CE8449

 10:20:22 PM IST

 HOTS

 Emergency

in the SPA inspection panel.

8 of 8 Services Available

 Search by facility name, block, or keywords...

All

Academic Support

Library Services

Transportation

Hostel Services

Health & Wellness

Technical Support

Security

Student Welfare

ACADEMIC

★ 4.8/5.0

Academic Support & FFCS Cell (TT)
Provides assistance for fee payment, course registration, and exam queries.

 Technology Tower, 1st Floor
 Mon-Fri 09:00-17:00

Click to inspect details →

LIBRARY

★ 4.6/5.0

Periyar EVR Central Library
Extensive collection of books, journals, and digital resources.

 Central Campus, Ground Floor
 Mon-Sat 08:00-22:00

Click to inspect details →

TRANSPORT

★ 4.3/5.0

Campus Transportation & Shuttles
Shuttle services connecting main campus, Katpadi and hostel zones.

 Transport Depot, 2nd Floor
 24/7

Click to inspect details →

HOSTEL

★ 4.5/5.0

Hostel Services & Chief Warden Office
Accommodation management, mess, and grievance redressal.

 Hostel Block A, Ground Floor
 Mon-Sat 07:00-20:00

Click to inspect details →

HEALTH

★ 4.7/5.0

VIT Health Centre & Emergency Care

IT

★ 4.4/5.0

Centre for Technical Support (CTS)

SECURITY

★ 4.2/5.0

Campus Security & Lost & Found

WELFARE

★ 4.6/5.0

Office of Student Welfare & Clubs

ACADEMIC

★ 4.8 / 5.0 Rating

×

Academic Support & FFCS Cell (TT)

Department Category: Academic Support

Provides assistance for fee payment, course registration, and exam queries.

CAMPUS LOCATION

Technology Tower, 1st Floor

OPERATING HOURS

Mon-Fri 09:00-17:00

OFFICER IN CHARGE

Dr. S. Kumar (Coordinator)

DIRECT PHONE / INTERCOM

+91-9795-111111

OFFICIAL EMAIL

tt@vit.ac.in

Available Key Facilities & Support:

✓ Online fee portal

✓ Live chat

✓ Counselling

Campus Assistance Hub

VIT VELLORE MAIN CAMPUS

LATEST NOTICE

Welcome to VIT Vellore Cam

AJAY BERLIN

248CE9449

10:20:24 PM IST

HOTS

Emergency

Live telemetry demonstrating genuine React ES6 Class Component behavior, constructor initialization, props consumption, lifecycle hooks (`componentDidMount` , `componentDidUpdate`), and Component API methods (`this.setState` , `this.forceUpdate`).

ACTIVE SERVICE PROP

None (Select Card)

TOTAL SERVICES PROP

8 Registered

INTERNAL STATE COUNTER

1

LAST RENDER TIMESTAMP

10:20:23 PM

⚡ Trigger this.setState()

🔧 Trigger this.forceUpdate()

🗑️ Clear Event Stream

Real-Time Component Life Cycle Log Stream

7 Events Logged

> [10:20:20 PM] [CONSTRUCTOR] Initialized state: counter=0, status=READY
> [10:20:20 PM] [LIFECYCLE: componentDidMount] Component mounted to DOM. Props received (totalServices: 8)
> [10:20:22 PM] [LIFECYCLE: componentDidUpdate] Active Service transitioned: "None" → "Academic Support & FFCS Cell (TT)"
> [10:20:23 PM] [LIFECYCLE: componentDidUpdate] Active Service transitioned: "Academic Support & FFCS Cell (TT)" → "None"
> [10:20:23 PM] [LIFECYCLE: componentDidUpdate] Active Service transitioned: "None" → "Periyar EVR Central Library"
> [10:20:23 PM] [LIFECYCLE: componentDidUpdate] Active Service transitioned: "Periyar EVR Central Library" → "None"
> [10:20:23 PM] [COMPONENT API: this.setState()] State updated! Counter incremented to 1

Vellore Institute of Technology

Campus Assistance Hub — Single Page Application (SPA)

OFFICIAL PORTALS

→ VTOP Portal

STUDENT SUBMISSION DETAILS

Student Name: AJAY BERLIN

Scalability Justification: 5 to 50 Campus Services

Question: Suppose the Campus Assistance Hub is expanded from 5 services to 50 services. Analyze how your component design would affect code reusability, duplication, and maintainability. Identify one important component–design decision made in your application and justify why it would remain useful as the application grows.

1. Impact on Reusability, Duplication & Maintainability

- **100% Component Reusability (Zero JSX Duplication):** Rather than hardcoding separate HTML cards for each service, the system encapsulates all layout, styling, badges, and click triggers in a single, reusable `<ServiceCard />` component. Whether rendering 5 or 50 services, React maps over the dataset dynamically:

```
{services.map(s => <ServiceCard key={s.id} service={s} onSelect={onSelect} />)}
```
- **Decoupled Data Architecture (\$O(1)\$ Code Overhead):** Expanding to 50 services requires adding records strictly to `src/data/services.js` (or fetching from a REST API). Zero component modifications or JSX rewrites are needed.
- **Centralized Maintainability (\$O(1)\$ Modification Cost):** If card styling, hover effects, or accessibility tags need updates, modifying `ServiceCard.jsx` immediately updates all 50 rendered cards uniformly across the entire application.

2. Key Component–Design Decision: Decoupled State Container Architecture

```
{services.map(s => <ServiceCard key={s.id} service={s} onSelect={onSelect} />)}
```

- **Decoupled Data Architecture (\$O(1)\$ Code Overhead):** Expanding to 50 services requires adding records strictly to `src/data/services.js` (or fetching from a REST API). Zero component modifications or JSX rewrites are needed.
- **Centralized Maintainability (\$O(1)\$ Modification Cost):** If card styling, hover effects, or accessibility tags need updates, modifying `ServiceCard.jsx` immediately updates all 50 rendered cards uniformly across the entire application.

2. Key Component–Design Decision: Decoupled State Container Architecture

The central architectural decision was **lifting state up to a top-level orchestrator (`App.jsx`) while maintaining presentational purity in card components**:

- **Single Source of Truth:** Dynamic filtering, search querying, and active modal selection are managed cleanly without prop drilling.
- **Virtualization & Pagination Ready:** As the catalog expands to 50+ services, a virtualization window (e.g. `react-window`) or paginated chunking can be dropped inside `ServiceGrid` without modifying the card component contract.
- **Predictable Life Cycle Hooks:** Selected services trigger `componentDidUpdate()` in the telemetry monitor cleanly, allowing audit trails, analytics, and dynamic DOM title updates.

Dimension	5 Services (Baseline)	50 Services (Scaled)	Architectural Justification
Component Instances	1 Reusable Card Definition	1 Reusable Card Definition	Zero code duplication; 100% component reuse
Maintenance Cost	Low	\$O(1)\$ Complexity	Only the JSON/data array expands

24/7 PRIORITY HELPLINES



VIT Vellore Emergency Assistance Directory

Direct priority contacts for health, security, hostel, and anti-ragging assistance.



Campus Ambulance & Emergency Clinic

24/7 on-campus ambulance with emergency tie-up to CMC Vellore.

+91-9795-555555 Ext. 5555

Call Now



Chief Security Control Room (Gate 1)

Main security hub, patrol dispatch, and gate access.

+91-9795-777777 Ext. 7777

Call Now



VIT Health Centre Helpline

Inpatient medical doctor on duty & dispensary.

+91-9795-555556 Ext. 5556

Call Now



Chief Warden Office (Hostel Emergency)

Men's & Ladies' Hostel supervisor desk.

+91-9795-444444 Ext. 4444

Call Now

+91-9795-555555 Ext. 5555



Chief Security Control Room (Gate 1)

Main security hub, patrol dispatch, and gate access.

+91-9795-777777 Ext. 7777

Call Now



VIT Health Centre Helpline

Inpatient medical doctor on duty & dispensary.

+91-9795-555556 Ext. 5556

Call Now



Chief Warden Office (Hostel Emergency)

Men's & Ladies' Hostel supervisor desk.

+91-9795-444444 Ext. 4444

Call Now



National Anti-Ragging Toll-Free Helpline

UGC & VIT zero-tolerance grievance monitoring.

1800-180-5522 Toll-Free

Call Now



Vellore Local Police Assistance

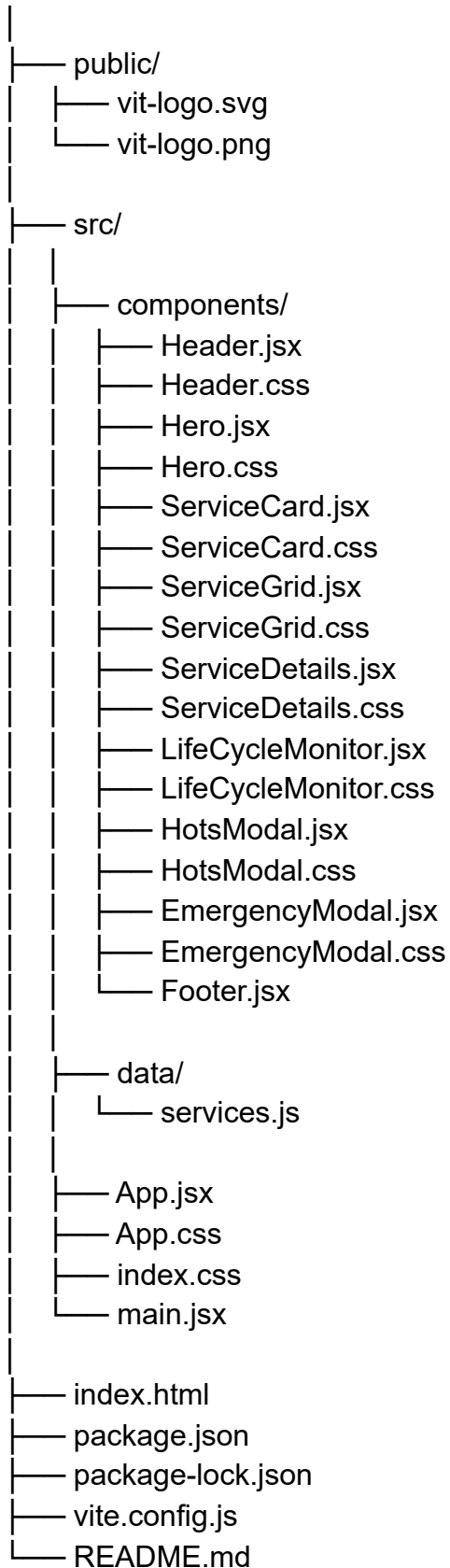
Katpadi Police Station jurisdictional support.

100 / 0416-222222 City Police

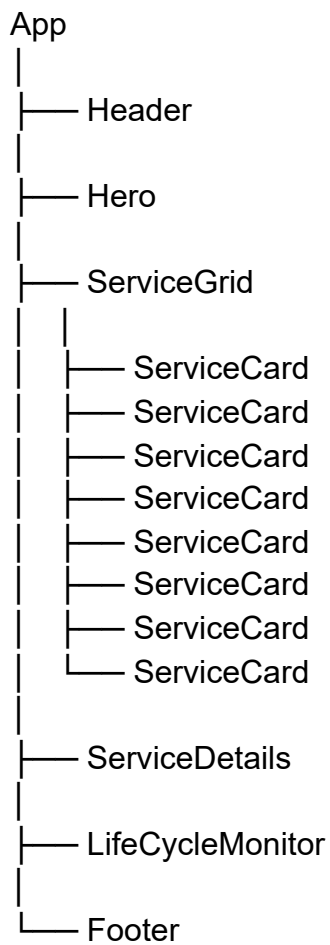
Call Now

PROJECT STRUCTURE

Campus Assistance Hub



Component Hierarchy



GITHUB REPOSITORY LINK:

<https://github.com/ajayberlin0608/Campus-Assistance-Hub>

TESTED IN INCOGNITO:

